

Solutions to Exercise 2B

Chapter 2: Fractions – Division & Reciprocals

Prepared for the Notebook Project

Exercise Overview

This solution guide covers **Exercise 2B** from the chapter on **Fractions**. It includes the step-by-step resolution of:

- **Question 1:** Finding the reciprocals of proper fractions, improper fractions, whole numbers, and mixed numbers.
- **Question 2:** Division of simple and mixed fractions with full cross-simplification.
- **Questions 3–12:** Comprehensive word problems translating real-world scenarios into fractional division equations.

All operations follow the rigorous mathematical principles of fractional arithmetic: converting mixed numbers to improper fractions, multiplying by the reciprocal of the divisor, and reducing terms by cross-cancelling prime factors.

1 Question 1: Find the Reciprocal of Fractions

Rule of Reciprocals: The reciprocal (or multiplicative inverse) of a fraction $\frac{a}{b}$ is $\frac{b}{a}$, since $\frac{a}{b} \times \frac{b}{a} = 1$. For any integer n , it can be written as $\frac{n}{1}$, and its reciprocal is $\frac{1}{n}$. For mixed numbers, we must first convert them to improper fractions before finding their reciprocals.

Detailed Solutions

(i) **Find the reciprocal of $\frac{13}{17}$:**

→ Since the fraction is already in the form $\frac{a}{b}$ with $a = 13$ and $b = 17$, we swap the numerator and denominator.

→ **Reciprocal:** $\frac{17}{13}$ (or $1\frac{4}{13}$).

(ii) **Find the reciprocal of $\frac{3}{313}$:**

→ We swap the numerator (3) and denominator (313).

→ **Reciprocal:** $\frac{313}{3}$ (or $104\frac{1}{3}$).

(iii) **Find the reciprocal of 217:**

→ The whole number 217 is written as $\frac{217}{1}$ in fractional form.

→ **Reciprocal:** $\frac{1}{217}$.

(iv) **Find the reciprocal of $\frac{1}{1024}$:**

→ Swapping the numerator and denominator gives $\frac{1024}{1}$.

→ **Reciprocal:** 1024.

(v) **Find the reciprocal of $3\frac{1}{3}$:**

→ Convert the mixed number to an improper fraction:

$$3\frac{1}{3} = \frac{(3 \times 3) + 1}{3} = \frac{10}{3}$$

→ Swapping the numerator and denominator of $\frac{10}{3}$ gives its reciprocal.

→ **Reciprocal:** $\frac{3}{10}$.

(vi) **Find the reciprocal of $11\frac{1}{11}$:**

→ Convert to an improper fraction:

$$11\frac{1}{11} = \frac{(11 \times 11) + 1}{11} = \frac{122}{11}$$

→ **Reciprocal:** $\frac{11}{122}$.

(vii) **Find the reciprocal of $1\frac{1}{2}$:**

→ Convert to an improper fraction:

$$1\frac{1}{2} = \frac{(1 \times 2) + 1}{2} = \frac{3}{2}$$

→ **Reciprocal:** $\frac{2}{3}$.

(viii) **Find the reciprocal of $125\frac{1}{8}$:**

→ Convert to an improper fraction:

$$125\frac{1}{8} = \frac{(125 \times 8) + 1}{8} = \frac{1000 + 1}{8} = \frac{1001}{8}$$

→ **Reciprocal:** $\frac{8}{1001}$.

2 Question 2: Division of Fractions

Rule of Division: To divide one fraction by another, we multiply the dividend (the first fraction) by the reciprocal of the divisor (the second fraction):

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{a \times d}{b \times c}$$

We perform cross-cancellation where possible to write the quotient in its simplest form.

Detailed Solutions

(i) **Divide** $\frac{1}{2} \div \frac{1}{3}$:

→ Multiply by the reciprocal of the divisor $\frac{1}{3}$ (which is 3):

$$\frac{1}{2} \div \frac{1}{3} = \frac{1}{2} \times 3 = \frac{3}{2} = 1\frac{1}{2}$$

(ii) **Divide** $\frac{3}{14} \div \frac{11}{42}$:

→ Multiply by the reciprocal of $\frac{11}{42}$ (which is $\frac{42}{11}$):

$$\frac{3}{14} \div \frac{11}{42} = \frac{3}{14} \times \frac{42}{11}$$

→ Cross-cancelling 42 and 14 (since $42 = 14 \times 3$):

$$\frac{3}{1} \times \frac{3}{11} = \frac{9}{11}$$

(iii) **Divide** $1 \div 6\frac{2}{3}$:

→ Convert $6\frac{2}{3}$ to an improper fraction: $6\frac{2}{3} = \frac{20}{3}$.

→ Multiply 1 by the reciprocal of $\frac{20}{3}$ (which is $\frac{3}{20}$):

$$1 \div \frac{20}{3} = 1 \times \frac{3}{20} = \frac{3}{20}$$

(iv) **Divide** $13\frac{1}{3} \div 10$:

→ Convert $13\frac{1}{3}$ to an improper fraction: $13\frac{1}{3} = \frac{40}{3}$.

→ Multiply by the reciprocal of 10 (which is $\frac{1}{10}$):

$$\frac{40}{3} \div 10 = \frac{40}{3} \times \frac{1}{10}$$

→ Cancelling the common factor of 10 from 40 and 10:

$$\frac{4}{3} \times \frac{1}{1} = \frac{4}{3} = 1\frac{1}{3}$$

(v) **Divide** $12\frac{1}{12} \div \frac{5}{36}$:

→ Convert $12\frac{1}{12}$ to an improper fraction: $12\frac{1}{12} = \frac{145}{12}$.

→ Multiply by the reciprocal of $\frac{5}{36}$ (which is $\frac{36}{5}$):

$$\frac{145}{12} \times \frac{36}{5}$$

→ Cross-cancelling:

– $36 \div 12 = 3$ (cancelling 12 and 36)

– $145 \div 5 = 29$ (cancelling 5 and 145)

$$29 \times 3 = \mathbf{87}$$

(vi) **Divide** $\frac{13}{46} \div 2\frac{1}{11}$:

→ Convert $2\frac{1}{11}$ to an improper fraction: $2\frac{1}{11} = \frac{23}{11}$.

→ Multiply by the reciprocal of $\frac{23}{11}$ (which is $\frac{11}{23}$):

$$\frac{13}{46} \times \frac{11}{23} = \frac{13 \times 11}{46 \times 23} = \frac{\mathbf{143}}{\mathbf{1058}}$$

(vii) **Divide** $\frac{20}{31} \div 54$:

→ Multiply by the reciprocal of 54 (which is $\frac{1}{54}$):

$$\frac{20}{31} \times \frac{1}{54}$$

→ Simplify by dividing 20 and 54 by their greatest common factor, 2:

$$\frac{10}{31} \times \frac{1}{27} = \frac{10}{31 \times 27} = \frac{\mathbf{10}}{\mathbf{837}}$$

(viii) **Divide** $9\frac{4}{5} \div 3\frac{23}{25}$:

→ Convert mixed numbers to improper fractions:

$$9\frac{4}{5} = \frac{49}{5} \quad \text{and} \quad 3\frac{23}{25} = \frac{(3 \times 25) + 23}{25} = \frac{75 + 23}{25} = \frac{98}{25}$$

→ Multiply $\frac{49}{5}$ by the reciprocal of $\frac{98}{25}$ (which is $\frac{25}{98}$):

$$\frac{49}{5} \times \frac{25}{98}$$

→ Cross-cancelling:

– 49 and 98 by dividing by 49 ($49 \div 49 = 1$ and $98 \div 49 = 2$)

– 25 and 5 by dividing by 5 ($25 \div 5 = 5$ and $5 \div 5 = 1$)

$$\frac{1}{1} \times \frac{5}{2} = \frac{5}{2} = \mathbf{2\frac{1}{2}}$$

(ix) **Divide** $32\frac{1}{2} \div 8\frac{3}{4}$:

→ Convert mixed numbers to improper fractions:

$$32\frac{1}{2} = \frac{65}{2} \quad \text{and} \quad 8\frac{3}{4} = \frac{35}{4}$$

→ Multiply $\frac{65}{2}$ by the reciprocal of $\frac{35}{4}$ (which is $\frac{4}{35}$):

$$\frac{65}{2} \times \frac{4}{35}$$

→ Cross-cancelling:

- 4 and 2 by dividing by 2 ($4 \div 2 = 2$ and $2 \div 2 = 1$)
- 65 and 35 by dividing by 5 ($65 \div 5 = 13$ and $35 \div 5 = 7$)

$$\frac{13}{1} \times \frac{2}{7} = \frac{26}{7} = 3\frac{5}{7}$$

(x) **Divide** $8\frac{1}{21} \div 1\frac{6}{7}$:

→ Convert mixed numbers to improper fractions:

$$8\frac{1}{21} = \frac{(8 \times 21) + 1}{21} = \frac{168 + 1}{21} = \frac{169}{21} \quad \text{and} \quad 1\frac{6}{7} = \frac{13}{7}$$

→ Multiply $\frac{169}{21}$ by the reciprocal of $\frac{13}{7}$ (which is $\frac{7}{13}$):

$$\frac{169}{21} \times \frac{7}{13}$$

→ Cross-cancelling:

- 169 and 13 by dividing by 13 ($169 \div 13 = 13$ and $13 \div 13 = 1$, since $13^2 = 169$)
- 7 and 21 by dividing by 7 ($7 \div 7 = 1$ and $21 \div 7 = 3$)

$$\frac{13}{3} \times \frac{1}{1} = \frac{13}{3} = 4\frac{1}{3}$$

(xi) **Divide** $6\frac{3}{16} \div 3\frac{1}{7}$:

→ Convert mixed numbers to improper fractions:

$$6\frac{3}{16} = \frac{99}{16} \quad \text{and} \quad 3\frac{1}{7} = \frac{22}{7}$$

→ Multiply $\frac{99}{16}$ by the reciprocal of $\frac{22}{7}$ (which is $\frac{7}{22}$):

$$\frac{99}{16} \times \frac{7}{22}$$

→ Cross-cancelling 99 and 22 by dividing by 11 ($99 \div 11 = 9$ and $22 \div 11 = 2$):

$$\frac{9}{16} \times \frac{7}{2} = \frac{63}{32} = 1\frac{31}{32}$$

(xii) **Divide** $3\frac{11}{15} \div 19\frac{3}{5}$:

→ Convert mixed numbers to improper fractions:

$$3\frac{11}{15} = \frac{56}{15} \quad \text{and} \quad 19\frac{3}{5} = \frac{(19 \times 5) + 3}{5} = \frac{95 + 3}{5} = \frac{98}{5}$$

→ Multiply $\frac{56}{15}$ by the reciprocal of $\frac{98}{5}$ (which is $\frac{5}{98}$):

$$\frac{56}{15} \times \frac{5}{98}$$

→ Cross-cancelling:

- 5 and 15 by dividing by 5 ($5 \div 5 = 1$ and $15 \div 5 = 3$)
- 56 and 98 by dividing by their greatest common factor 14 ($56 \div 14 = 4$ and $98 \div 14 = 7$)

$$\frac{4}{3} \times \frac{1}{7} = \frac{4}{21}$$

3 Section 3: Practical Word Problems

This section covers the translation of real-world scenarios into fraction arithmetic equations, with a strong focus on division and scaling factors.

Question 3: Multiplication Multipliers

Problem: By what fraction should $\frac{9}{35}$ be multiplied to get $\frac{7}{15}$?

Solution:

Let the required fraction be x . According to the problem statement, we have:

$$\frac{9}{35} \times x = \frac{7}{15}$$

To isolate x , we divide both sides by $\frac{9}{35}$, which is equivalent to multiplying by its reciprocal $\frac{35}{9}$:

$$x = \frac{7}{15} \div \frac{9}{35} = \frac{7}{15} \times \frac{35}{9}$$

We cross-simplify 35 and 15 by dividing both by their common factor of 5:

$$15 \div 5 = 3$$

$$35 \div 5 = 7$$

Substituting these simplified values:

$$x = \frac{7}{3} \times \frac{7}{9} = \frac{7 \times 7}{3 \times 9} = \frac{49}{27}$$

Convert the improper fraction $\frac{49}{27}$ to a mixed number:

$$\frac{49}{27} = 1\frac{22}{27}$$

Therefore, the fraction should be multiplied by $1\frac{22}{27}$.

Question 4: Unit Cost of Cloth

Problem: If the cost of $6\frac{1}{4}$ m of cloth is Rs. $1421\frac{7}{8}$, then find the cost of 1 metre of cloth.

Solution:

Let the cost of 1 metre of cloth be C . We are given:

- Total Length of Cloth = $6\frac{1}{4}$ m = $\frac{25}{4}$ m
- Total Cost of Cloth = Rs. $1421\frac{7}{8}$ = Rs. $\frac{(1421 \times 8) + 7}{8}$ = Rs. $\frac{11368 + 7}{8}$ = Rs. $\frac{11375}{8}$

Using the unitary method, the cost per metre is:

$$C = \text{Total Cost} \div \text{Total Length} = \frac{11375}{8} \div \frac{25}{4} = \frac{11375}{8} \times \frac{4}{25}$$

We cross-simplify:

- Simplify the denominators: $8 \div 4 = 2$ and $4 \div 4 = 1$.
- Divide 11375 by 25: $11375 \div 25 = 455$.

Substituting these simplified values:

$$C = \frac{455}{2} \times \frac{1}{1} = \text{Rs. } 227\frac{1}{2} \quad (\text{or Rs. } 227.50)$$

Therefore, the cost of 1 metre of cloth is **Rs. $227\frac{1}{2}$** .

Question 5: Cutting Wall Paper Pieces

Problem: How many pieces of length $3\frac{5}{7}$ m each can be cut from a wall paper 52 m long?

Solution:

We are given:

- Total length of wall paper = 52 m
- Length of one piece = $3\frac{5}{7}$ m = $\frac{(3 \times 7) + 5}{7}$ m = $\frac{26}{7}$ m

Let N be the total number of pieces. To find N , we divide the total length by the length of each piece:

$$N = 52 \div \frac{26}{7} = 52 \times \frac{7}{26}$$

Notice that 52 is exactly double of 26 ($52 \div 26 = 2$):

$$N = 2 \times 7 = 14$$

Therefore, exactly 14 pieces can be cut from the wall paper.

Question 6: Division of a Log of Wood

Problem: A log of wood $6\frac{2}{7}$ m in length was cut into 11 pieces of equal length. What is the length of each piece?

Solution:

We are given:

- Total length of the log = $6\frac{2}{7}$ m = $\frac{(6 \times 7) + 2}{7}$ m = $\frac{44}{7}$ m
- Number of equal pieces = 11

Let L be the length of each piece. We divide the total length by the number of pieces:

$$L = \frac{44}{7} \div 11 = \frac{44}{7} \times \frac{1}{11}$$

Cross-simplifying 44 and 11 (since $44 \div 11 = 4$):

$$L = \frac{4}{7} \times 1 = \frac{4}{7} \text{ m}$$

Therefore, the length of each piece of wood is $\frac{4}{7}$ m.

Question 7: Fuel Efficiency and Distance

Problem: A car covers $641\frac{1}{4}$ km in $43\frac{1}{8}$ litres of fuel. How much distance can this car cover in 1 litre of fuel?

Solution:

We are given:

- Total Distance Covered = $641\frac{1}{4}$ km = $\frac{(641 \times 4) + 1}{4}$ km = $\frac{2564 + 1}{4}$ km = $\frac{2565}{4}$ km
- Total Fuel Consumed = $43\frac{1}{8}$ litres = $\frac{(43 \times 8) + 1}{8}$ litres = $\frac{344 + 1}{8}$ litres = $\frac{345}{8}$ litres

To find the distance covered per litre of fuel (D), we divide total distance by total fuel:

$$D = \text{Total Distance} \div \text{Total Fuel} = \frac{2565}{4} \div \frac{345}{8} = \frac{2565}{4} \times \frac{8}{345}$$

We cross-simplify:

- Simplify the denominators: $8 \div 4 = 2$ and $4 \div 4 = 1$.
- Divide 2565 and 345 by their greatest common factor, 15:

$$2565 \div 15 = 171 \quad \text{and} \quad 345 \div 15 = 23$$

Substituting these simplified fractions back into the equation:

$$D = \frac{171}{1} \times \frac{2}{23} = \frac{342}{23} \text{ km}$$

Converting the improper fraction $\frac{342}{23}$ to a mixed number:

$$\frac{342}{23} = 14\frac{20}{23} \text{ km}$$

Therefore, the car can cover $14\frac{20}{23}$ **km** in 1 litre of fuel.

Question 8: Length of a Rectangular Plot

Problem: The area of a rectangular plot of land is $817\frac{4}{5}$ sq. m. If its breadth is $21\frac{3}{4}$ m, find its length.

Solution:

The area of a rectangle is given by the formula:

$$\text{Area} = \text{Length} \times \text{Breadth}$$

To find the length (W), we divide the Area by the Breadth:

$$\text{Length} = \text{Area} \div \text{Breadth}$$

We are given:

- Area = $817\frac{4}{5} \text{ m}^2 = \frac{(817 \times 5) + 4}{5} \text{ m}^2 = \frac{4085 + 4}{5} \text{ m}^2 = \frac{4089}{5} \text{ m}^2$
- Breadth = $21\frac{3}{4} \text{ m} = \frac{(21 \times 4) + 3}{4} \text{ m} = \frac{84 + 3}{4} \text{ m} = \frac{87}{4} \text{ m}$

Now, calculate the Length:

$$\text{Length} = \frac{4089}{5} \div \frac{87}{4} = \frac{4089}{5} \times \frac{4}{87}$$

Let's simplify by dividing 4089 and 87 by their common factor of 3:

$$4089 \div 3 = 1363 \quad \text{and} \quad 87 \div 3 = 29$$

Substituting these values:

$$\text{Length} = \frac{1363}{5} \times \frac{4}{29}$$

Next, check if 1363 is divisible by 29:

$$1363 \div 29 = 47 \quad (\text{since } 47 \times 29 = 1363)$$

Thus, the equation simplifies to:

$$\text{Length} = \frac{47}{5} \times \frac{4}{1} = \frac{188}{5} = 37\frac{3}{5} \text{ m}$$

Therefore, the length of the rectangular plot is $37\frac{3}{5}$ m.

Question 9: Width of a Sheet of Paper

Problem: The area of a sheet of paper is $623\frac{7}{10}$ sq. cm. If its length is $29\frac{7}{10}$ cm, find its width.

Solution:

Using the rectangle area formula (Width = Area $\div$ Length):

- Area = $623\frac{7}{10}$ cm² = $\frac{6237}{10}$ cm²
- Length = $29\frac{7}{10}$ cm = $\frac{297}{10}$ cm

Calculate the width (w):

$$w = \frac{6237}{10} \div \frac{297}{10} = \frac{6237}{10} \times \frac{10}{297}$$

The denominators (10) cancel out:

$$w = \frac{6237}{297}$$

Dividing 6237 by 297 directly:

$$6237 \div 297 = 21 \quad (\text{since } 21 \times 297 = 6237)$$

Therefore, the width of the sheet of paper is exactly 21 **cm**.

Question 10: Net Weight of a Single Sheet

Problem: A bundle of 500 sheets of paper has a net weight of $2\frac{3}{10}$ kg. Find the weight of 1 sheet of paper.

Solution:

We are given:

- Total number of sheets = 500
- Net weight of the bundle = $2\frac{3}{10}$ kg = $\frac{23}{10}$ kg

Let W be the weight of 1 sheet of paper. We divide the total weight by the number of sheets:

$$W = \text{Total Weight} \div 500 = \frac{23}{10} \div 500 = \frac{23}{10} \times \frac{1}{500} = \frac{23}{5000} \text{ kg}$$

To express this in a more practical unit (grams), we multiply by 1000 (since 1 kg = 1000 g):

$$w = \frac{23}{5000} \times 1000 \text{ g} = \frac{23}{5} \text{ g} = 4\frac{3}{5} \text{ g} \quad (\text{or } 4.6 \text{ g})$$

Therefore, the weight of a single sheet of paper is $\frac{23}{5000}$ **kg** (or 4.6 g).

Question 11: Speed of a Car

Problem: A car travels $283\frac{1}{2}$ km in $4\frac{2}{3}$ hours. How far does this car go in 1 hour, travelling at the same speed?

Solution:

The distance traveled in 1 hour is equivalent to the car's uniform speed (S):

$$\text{Speed } S = \text{Total Distance} \div \text{Total Time}$$

We are given:

- Total Distance = $283\frac{1}{2}$ km = $\frac{(283 \times 2) + 1}{2}$ km = $\frac{566 + 1}{2}$ km = $\frac{567}{2}$ km
- Total Time = $4\frac{2}{3}$ hours = $\frac{(4 \times 3) + 2}{3}$ hours = $\frac{12 + 2}{3}$ hours = $\frac{14}{3}$ hours

Now, calculate the speed (S):

$$S = \frac{567}{2} \div \frac{14}{3} = \frac{567}{2} \times \frac{3}{14}$$

Let's simplify by checking if 567 is divisible by 14's prime factors (2 and 7). Since 567 is odd, we check divisibility by 7:

$$567 \div 7 = 81 \quad \text{and} \quad 14 \div 7 = 2$$

Substituting these values:

$$S = \frac{81}{2} \times \frac{3}{2} = \frac{81 \times 3}{2 \times 2} = \frac{243}{4} = 60\frac{3}{4} \text{ km/h}$$

Therefore, travelling at the same speed, the car goes $60\frac{3}{4}$ **km** in 1 hour.

Question 12: Finding the Other Fraction

Problem: The product of two fractions is $8\frac{7}{25}$. If one of them is $3\frac{1}{15}$, find the other fraction.

Solution:

Let the two fractions be A and B , where:

$$A \times B = \text{Product}$$

We are given:

- Product $= 8\frac{7}{25} = \frac{(8 \times 25) + 7}{25} = \frac{200 + 7}{25} = \frac{207}{25}$
- One fraction $A = 3\frac{1}{15} = \frac{(3 \times 15) + 1}{15} = \frac{45 + 1}{15} = \frac{46}{15}$

To find the other fraction (B), we divide the Product by A :

$$B = \text{Product} \div A = \frac{207}{25} \div \frac{46}{15} = \frac{207}{25} \times \frac{15}{46}$$

We cross-simplify:

- Simplify 15 and 25 by dividing by 5: $15 \div 5 = 3$ and $25 \div 5 = 5$.
- Simplify 207 and 46 by dividing by their greatest common factor, 23:

$$207 \div 23 = 9 \quad \text{and} \quad 46 \div 23 = 2$$

Substituting these simplified fractions:

$$B = \frac{9}{5} \times \frac{3}{2} = \frac{9 \times 3}{5 \times 2} = \frac{27}{10} = 2\frac{7}{10}$$

Therefore, the other fraction is $2\frac{7}{10}$.